SWE 4638

HTML Fundamentals

Building the Structure of the Web

Web and Mobile Application Development • Lecture 1

What is HTML?

HyperText Markup Language
The standard language for creating and structuring content on web pages. It defines the meaning and structure of content — not its appearance.

Structure

Defines headings, paragraphs, lists, links, images

Semantic

Tags describe the meaning of content, not its look

Universal

Understood by every web browser on every device

HTML Document Structure

```html
<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Hello World</h1>
    <p>Welcome!</p>
  </body>
</html>
```

DOCTYPE

Tells browser this is HTML5

<head>

Metadata, title, linked stylesheets — not visible on page

<body>

All visible content goes here — text, images, links

Essential HTML Tags: Text

Tag	Purpose
`<h1> – <h6>`	Headings (h1 = largest, h6 = smallest)
`<p>`	Paragraph — a block of text
` `	Line break (self-closing, no end tag)
`<strong>`	Bold / important text
`<em>`	Italic / emphasized text
`<span>`	Inline container for styling a portion of text
`<blockquote>`	Long quotation — indented block

Structuring Content

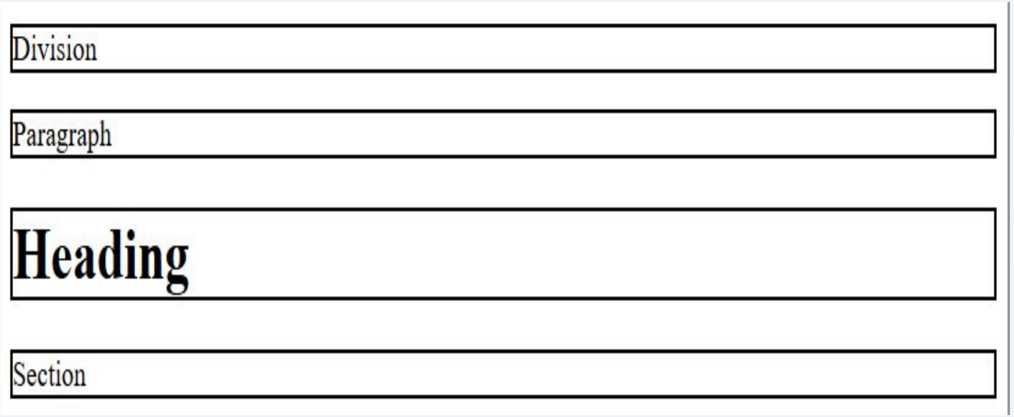
Lists, Links, Images & Semantic Elements

HTML Block and Inline Elements

What are Block Elements?

- Take full width of the parent container
- Always start on a new line
- Can contain inline and other block elements

```html
<div>Division</div>
<p>Paragraph</p>
<h1>Heading</h1>
<section>Section</section>
```



HTML Block and Inline Elements

What are Inline Elements?

- Take only necessary width
- Do NOT start on a new line
- Usually used inside block elements

```
<span>Text</span>
<a href=”#”>Link</a>
<strong>Bold</strong>
<em>Italic</em>
```

Key Attributes

Class

- Space-separated list of class identifiers.
- Represents all the classes that this element belongs to.
- Heavily used in CSS and JS for matching portions of contents

Id

- Represents an element’s unique identifier
- Must be unique within this specific HTML document
- Heavily used in CSS and JS for finding/matching this specific element

HTML Lists

Unordered List

```html
<ul>
  <li>Item One</li>
  <li>Item Two</li>
  <li>Item Three</li>
</ul>
```

Result: • Item One • Item Two • Item Three

Ordered List

```html
<ol>
  <li>First Step</li>
  <li>Second Step</li>
  <li>Third Step</li>
</ol>
```

Result: 1. First Step 2. Second Step 3. Third Step

Definition List: <dl> with <dt> (term) and <dd> (description) — great for glossaries and FAQ sections.

Links & Images

Hyperlinks <a>

```
<a href="url">Link Text</a>
```

Key Attributes:

- href — the destination URL
- target="_blank" — open in new tab
- Relative paths: ./about.html
- Anchor links: #section-id

Images <img>

```
<img src="photo.jpg"
     alt="Description"
     width="400">
```

Key Points:

- Self-closing tag (no </img>)
- alt is required for accessibility
- Formats: JPG, PNG, GIF, SVG, WebP
- Use width/height or CSS to resize

HTML Tables

```html
<table>
  <thead>
    <tr>
      <th>Name</th>
      <th>Role</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Alex</td>
      <td>Trainer</td>
    </tr>
  </tbody>
</table>
```

Rendered Result:

Name	Role
Alex	Trainer

Key Elements:

- <table> — container
- <tr> — table row
- <th> — header cell (bold, centered)
- <td> — data cell
- colspan / rowspan — merge cells

HTML Table

<table>
<tr><td colspan="2">Cell 1</td><td>Cell 2</td><td>Cell 3</td></tr>
<tr><td rowspan="3">Cell 4</td><td>Cell 5</td><td>Cell 6</td><td>Cell 7</td></tr>
<tr><td colspan="3">Cell 8</td></tr>
<tr><td>Cell 9</td><td colspan="2">Cell 10</td></tr>
</table>

HTML Forms

Forms collect user input — the backbone of interactive web applications. Every login, search bar, and registration uses <form>.

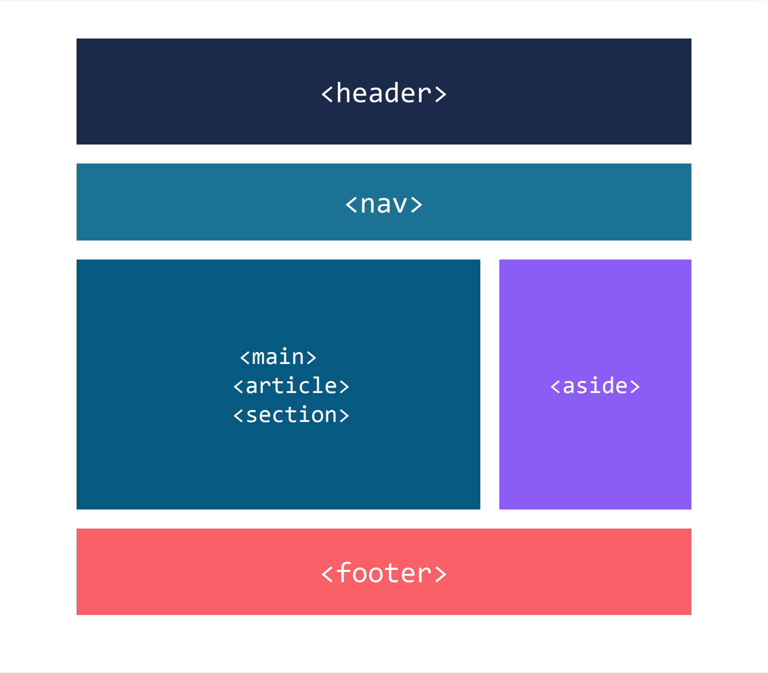
```html
<form action="/submit" method="POST">

  <label for="name">Name:</label>
  <input type="text" id="name"
         name="name" required>

  <select name="level">
    <option>Beginner</option>
    <option>Advanced</option>
  </select>

  <button type="submit">Go</button>
```

Common Input Types

- text — single-line text
- email — email with validation
- password — hidden characters
- number — numeric input
- date — date picker
- checkbox — multiple selections
- radio — single selection
- file — file upload
- textarea — multi-line text

Semantic HTML5 Elements

Semantic tags describe what the content IS — not how it looks. They improve accessibility, SEO, and code readability.



<header>	Introductory content or navigation
<nav>	Navigation links
<main>	Primary content (one per page)
<article>	Self-contained composition
<section>	Thematic grouping of content
<aside>	Sidebar / related content
<footer>	Footer info, copyright, links

<div> vs Semantic Tags

❌ Non-semantic (Div Soup)

```html
<div class="header">
  <div class="nav">...</div>
</div>
<div class="content">
  <div class="article">
    ...
  </div>
</div>
<div class="footer">...</div>
```

✓ Semantic HTML5

```html
<header>
  <nav>...</nav>
</header>
<main>
  <article>
    ...
  </article>
</main>
<footer>...</footer>
```

HTML5 Form Attributes & Validation

Attribute	Example	What it Does
`required`	`<input required>`	Field must be filled before submit
`placeholder`	`placeholder="Your name"`	Ghost text shown when field is empty
`maxlength`	`maxlength="200"`	Limits number of characters
`min / max`	`min="1" max="10"`	Range for number/date inputs
`pattern`	`pattern="[0-9]{3}"`	Regex pattern for validation
`disabled`	`<input disabled>`	Grays out the field, can't interact

HTML Best Practices

Always declare DOCTYPE

<!DOCTYPE html> ensures standards mode rendering

Use semantic elements

Prefer <nav>, <main>, <article> over generic <div>

Write meaningful alt text

Describe the image: alt="Golden retriever in park"

Keep it accessible

Use labels for form inputs, logical heading hierarchy

Validate your HTML

Use W3C Validator to catch errors early

Indent consistently

2 or 4 spaces — pick one and stick with it

Summary

- HTML provides the structure and meaning of web content
- Semantic tags improve accessibility and SEO
- Forms enable user interaction and data collection
- Tables organize data in rows and columns
- HTML is the foundation — CSS and JavaScript build on top

Next Lecture: CSS — Styling the Web